Twisted States in Higgs Boson Decay into Fermion Antifermion Pair

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1 Introduction

Problem Statement

Consider the decay of the Higgs boson into a pair of fermions $f\bar{f}$. The quantum state of the fermions is given by:

$$|f\bar{f}\rangle = \int \frac{d\vec{p}_1}{(2\pi)^3\sqrt{2E_1}} \frac{d\vec{p}_2}{(2\pi)^3\sqrt{2E_2}} \psi(\vec{p}_1, s_1; \vec{p}_2, s_2)|f(\vec{p}_1, s_1), \bar{f}(\vec{p}_2, s_2)\rangle,$$

where ψ is the two-particle wave function of fermions in momentum representation, s_i are the spin 4-vectors. In the rest frame of the fermion $s = (0, \boldsymbol{\xi})$.

1. Find ψ using the matrix element:

$$\langle f(\vec{p}_1, s_1), \bar{f}(\vec{p}_2, s_2)|S - 1|\text{Higgs boson}\rangle.$$

2. Now one can calculate the wave function in coordinate representation:

$$\psi(\vec{x}_1, \vec{x}_2) = \int \frac{d\vec{p}_1}{(2\pi)^3} \frac{d\vec{p}_2}{(2\pi)^3} \widetilde{\psi}(\vec{p}_1, s_1; \vec{p}_2, s_2) e^{+i(\vec{p}_1\vec{x}_1 + \vec{p}_2\vec{x}_2)}.$$

Show that $\psi(\vec{x}_1, \vec{x}_2)$ depends on the coordinate difference $\vec{r} = \vec{x}_1 - \vec{x}_2$ and calculate the wave function $\psi(\vec{r})$.

3. Expand the wave function in a series of spherical harmonics and investigate their behavior as a function of $\boldsymbol{\xi}_1 \cdot \hat{\vec{r}}, \boldsymbol{\xi}_2 \cdot \hat{\vec{r}}, \boldsymbol{\xi}_1 \cdot \boldsymbol{\xi}_2$. Give an interpretation.

2 Find wavefunction

Using the Fourier transform

$$\begin{aligned}
\psi\left(\vec{r}_1, \vec{r}_2\right) &= \int \frac{d^3p_1 d^3p_2 \widetilde{\psi}\left(\vec{p}_1 \vec{p}_2\right)}{(2\pi)^6 \sqrt{2E_p r_1)}} e^{i(\ldots)} \\
&= \int \frac{d^2p_1 d^2p_2}{(2\pi)^3} \langle f\overline{f} \mid S - 1\rangle e^{i(\ldots)} \\
&= \int \frac{d^3p}{(2\pi^3)} \frac{1}{2E_p} \frac{m_f}{v} \overline{u}(\vec{p_1}, \vec{s_1}) u(-\vec{p_2}, \vec{s_2}) e^{i(\ldots)}
\end{aligned}$$

$$S \mid i\rangle = \mid f\rangle = \ldots + \int \frac{d^3p_1 d^3p_2}{(2\pi)^6} \frac{\widetilde{\psi}\left(\vec{p_1}, \vec{p_2}\right)}{\sqrt{2E_1 2E_2}} \mid \vec{p_1}, \vec{s_1}; \vec{p_2}, \vec{s_2}\rangle + \ldots$$

$$\langle \vec{p}_1\vec{p}_2 \mid \vec{p}_1\vec{p}_2\rangle = (2\pi)^3\sqrt{2E_1 2E_2}\left[\delta^3\left(\vec{p}_1-\vec{p}_2\right)\delta'\left(\vec{p}_2-\vec{p}_1'\right) - \delta^3\left(\vec{p}_1-\vec{p}_2\right)\delta'\left(\vec{p}_2-\vec{p}_1'\right)\right]$$

Momentum

$$\begin{aligned} \mid p\rangle &\equiv \sqrt{2E_p}a_p^\dagger \mid 0\rangle \\ \langle \vec{q} \mid \vec{p}\rangle &= \langle \vec{q} \mid a_q\sqrt{2E_q}\sqrt{2E_p}a_p^\dagger \mid 0\rangle = 2E_p\left(2\pi\right)^3\delta^3(\vec{q}-\vec{p}) \\ a_q a_p^\dagger &= -a_q^\dagger a_p + \underbrace{\left\{a_q, a_p^\dagger\right\}}_{(2\pi^3)\delta^3(\vec{q}-\vec{p})} \end{aligned}$$

We'll get the following norm

$$\langle \vec{q}, s \mid \vec{p}, r\rangle = 2E_p\left(2\pi\right)^3\delta^3\left(\vec{q}-\vec{p}\right)\delta_{sr}$$

Substitute in our decay

$$\begin{aligned} \langle q_1, r_1; q_2, r_2 \mid p_1 s_1; p_2 s_2\rangle &= \sqrt{2E_{p_1}2E_{p_2}2E_{q_1}2E_{q_2}}\langle 0 \mid a_{q_1r_1}a_{q_2r_2}a_{p_1s_1}^\dagger a_{p_2s_2}^\dagger \mid 0\rangle \\ &= \sqrt{2E_{p_1}2E_{p_2}2E_{q_1}2E_{q_2}}\langle 0 \mid a_{q_1r_1}\left(-a_{q_2r_2}^\dagger a_{p_1s_1} + \left\{a_{q_2r_2}, a_{p_1s_1}^\dagger\right\}\right)a_{p_2s_2}^\dagger \mid 0\rangle \\ &= \left\{a_{q_2r_2}, a_{p_1s_1}^\dagger\right\}\langle 0 \mid a_{q_1r_1}a_{p_2s_2}^\dagger \mid 0\rangle - \langle 0 \mid a_{q_1r_1}a_{q_2r_2}^\dagger a_{p_1s_1}a_{p_2s_2}^\dagger \mid 0\rangle \\ &= \left\{a_{q_2r_2}, a_{p_1s_1}^\dagger\right\}\left\{a_{q_1r_1}, a_{p_2s_2}^\dagger\right\} - \left\{a_{q_1r_1}, a_{q_2r_2}^\dagger\right\}\left\{a_{p_1s_1}, a_{p_2s_2}^\dagger\right\} \end{aligned}$$

Finally will get

$$\begin{aligned} \langle \vec{q}_1 r_1; \vec{q}_2 r_2 \mid S \mid H\rangle &= \int \frac{d^3p_1 d^3p_2}{(2\pi)^6\sqrt{2E_{p_1}2E_{p_2}}}\widetilde{\psi}(\vec{p_1}, \vec{p_2})\langle q_1 r_1; q_2 r_2 \mid p_1 s_1; p_2 s_2\rangle \\ &= \int \frac{d^3p_1 d^3p_2}{\left(2\pi\right)^6\sqrt{2E_{p_1}2E_{p_2}}}\widetilde{\psi}(\vec{p_1}, \vec{p_2})\left[\left(2\pi\right)^6\delta^3(q_2-p_1)\delta_{r_2s_1}\delta^3(q_1-p_1)\delta_{r_2s_1} - \left(2\pi\right)^6\delta^3(q_2-p_2)\right. \\ &= \frac{\widetilde{\psi}(q_2, q_1)}{\sqrt{2E_{q_1}2E_{q_2}}}\delta_{r_2s_1}\delta_{r_1s_2} - \frac{\widetilde{\psi}(q_1, q_2)}{\sqrt{2E_{q_1}2E_{q_2}}}\delta_{r_1s_1}\delta_{r_2s_2} \end{aligned}$$

??

$$\frac{\widetilde{\psi}(q_2r_2; q_1r_1) - \widetilde{\psi}(q_1r_1; q_2r_2)}{\sqrt{2E_{q_1}2E_{q_2}}}$$

$$\widetilde{\psi}(q_1r_1; q_2r_2) \sim \sqrt{2E_{q_1}2E_{q_2}}\langle q_1r_1; q_2r_2 \mid S-1 \mid H\rangle$$

$$\begin{aligned} \widetilde{\psi}(\vec{r_1}s_1; \vec{r_2}s_2) &= \int \frac{d^3p_1 d^3p_2}{(2\pi)^6\sqrt{2E_{p1}2E_{p2}}}\widetilde{\psi}e^{i\vec{p_1}\cdot\vec{r_1}+i\vec{p_1}\vec{r_2}} \\ &= \int \frac{d^3p_1 d^3p_2}{(2\pi)^6}\underbrace{\langle p_1 s_1; p_2 s_2 \mid S-1 \mid H\rangle}_{(2\pi)^4\delta^4(P_H-P_1-P_2)i\frac{m_f}{v}\overline{u}(p_1s_1)v(p_2s_2)} e^{i\vec{p_1}\cdot\vec{r_1}+i\vec{p_1}\vec{r_2}} \end{aligned}$$

Using the spherical system of coordinates

$$\int \frac{d^3p_1 d^3p_2}{(2\pi)^6}\delta^4\left(P_H - E_{p_1} - E_{p_2}\right) i\frac{m_f}{v}\overline{u}(p_1 s_1)v(p_2 s_2)\Big|_{\vec{p_2}=-\vec{p_1}} e^{i\vec{p_1}\cdot\vec{r_1}+i\vec{p_1}\vec{r_2}}$$

$$d^3P_1 = \underbrace{P_1^2 dP_1}_{p_1 E_1 dE_1} d\Omega_{p_1}$$

$$E^2 = p^2 + m^2, \quad EdE = pdp$$

$$p_1^2 dp_1 = p_1 p_1 dp_1 = p_1 E_1 dE_1 = \sqrt{E_1^2 - m^2}E_1 dE_1$$

Using the property of delta function: $\int f(x)\delta(g(x)) = \sum_i \frac{f(x_i)}{|g'(x_i)|}$ and $g \equiv m_H - 2E_1 \equiv g(E_1)$, $g'(E_1) = -2$

$$\begin{aligned}&\frac{1}{(2\pi)^6}\int d\Omega dE_1 \sqrt{E_1^2 - m^2}E_1 \delta^4(m_H - 2E_1) i\frac{m_f}{v}\overline{u}(\vec{p}_1, s_1)v(-\vec{p}_1, s_2)\\ &= \frac{1}{(2\pi)^6}\frac{1}{2}\int d\Omega \sqrt{(\frac{m_H}{2})^2 - m^2}\frac{m_H}{2} i\frac{m_f}{v}\overline{u}(\vec{p}_1, s_1)v(-\vec{p}_1, s_2)\Big|_{|\vec{p}_1|=\sqrt{\left(\frac{m_H}{2}\right)^2 - m^2}}\end{aligned}$$

We have reached this expression

$$\psi(\vec{r}_1 - \vec{r}_2; s_1, s_2) \sim \int d\theta \sin\theta d\phi \underbrace{f(\theta,\phi)}_{\overline{u}v} e^{i|\vec{p}|\underbrace{(\vec{r}_1 - \vec{r}_2)}_{\equiv\vec{r}}\cdot\hat{n}}$$

where $\hat{n} = \frac{\vec{p}}{|\vec{p}|}$

3 Fermion Antifermion product

From Dirac's equation for fermion

$$u(p) = \begin{pmatrix}\sqrt{E_p + m_f}\chi(\vec{s_1})\\ \sqrt{E_p - m_f}(\overline{\sigma}\cdot\hat{n}_p)\chi(\vec{s_1})\end{pmatrix}$$

and antifermion $v(p,s) = c\gamma^0 u^*(p,s)$

$$v(-p) = \begin{pmatrix}\sqrt{E_p - m_f}(\overline{\sigma}\cdot\hat{n}_{-p})\eta(\vec{s_2})\\ \sqrt{E_p + m_f}\eta(\vec{s_2})\end{pmatrix}$$

taking in consideration $\hat{n}_{-p} = -\hat{n}_p$ and $\eta\hat{s}_2 = i\sigma_y\chi^*(\hat{s}_2)$

And for

$$\chi(\vec{s_1}) = \begin{pmatrix} \cos\frac{\theta_1}{2} \\ \sin\frac{\theta_1}{2} e^{i\phi_1} \end{pmatrix}$$

we get (keeping in mind that $\boldsymbol{p}_1 \upuparrows \boldsymbol{s}_1$ and $\boldsymbol{p}_2 \uparrow\downarrow \boldsymbol{s}_2$ we get the following substitution $(\theta_2, \phi_2) \to (\pi - \theta_2, \phi_2 + \pi)$)

$$\eta(\boldsymbol{s}_2) = i\sigma_y \chi^*(\boldsymbol{s}_2) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} \cos\frac{\pi-\theta_2}{2} \\ \sin\frac{\pi-\theta_2}{2} e^{-i(\phi_2+\pi)} \end{pmatrix} = \begin{pmatrix} -\cos\frac{\theta_2}{2} e^{-i\phi_2} \\ -\sin\frac{\theta_2}{2} \end{pmatrix}$$

Substitute in our

$$\overline{u} = u^\dagger \gamma^0 = (\sqrt{E_p + m_f}\chi^\dagger, -\sqrt{E_p - m_f}\chi^\dagger(\overline{\sigma} \cdot \hat{n}_p))$$

$$v = \begin{pmatrix} \sqrt{E_p - m_f}(\overline{\sigma} \cdot \hat{n}_{-p})\eta(\vec{s_2}) \\ \sqrt{E_p + m_f}\eta(\vec{s_2}) \end{pmatrix} = \begin{pmatrix} -\sqrt{E_p - m_f}(\overline{\sigma} \cdot \hat{n}_p)\eta(\vec{s_2}) \\ \sqrt{E_p + m_f}\eta(\vec{s_2}) \end{pmatrix}$$

$$\overline{u}v = -2\sqrt{(E_p + m_f)(E_p - m_f)}\chi^\dagger(\overline{\sigma} \cdot \hat{n}_p)\eta$$

$$\chi^\dagger(\boldsymbol{s}_1) = (\cos\frac{\theta_1}{2}, \sin\frac{\theta_1}{2} e^{-i\phi_1})$$

$$\eta(\boldsymbol{s}_2) = \begin{pmatrix} -\cos\frac{\theta_2}{2} e^{-i\phi_2} \\ -\sin\frac{\theta_2}{2} \end{pmatrix}$$

Expression for

$$\overline{\sigma} \cdot \hat{n}_p = \sigma_x n_x + \sigma_y n_y + \sigma_z n_z \quad = \begin{pmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{pmatrix}$$

$$\hat{n}_p = (\sin\theta_p \cos\phi_p, \sin\theta_p \sin\phi_p, \cos\theta_p)$$

$$\overline{s}_i = (\sin\theta_i \cos\phi_i, \sin\theta_i \sin\phi_i, \cos\theta_i)$$

$$\overline{\sigma} \cdot \hat{n}_p = \begin{pmatrix} \cos\theta_p & \sin\theta_p e^{-i\phi_p} \\ \sin\theta_p e^{i\phi_p} & -\cos\theta_p \end{pmatrix} = \begin{pmatrix} \cos\theta_p & \sin\theta_p e^{-i\phi_p} \\ \sin\theta_p e^{i\phi_p} & -\cos\theta_p \end{pmatrix}$$

$$(\boldsymbol{\sigma} \cdot \hat{n}_p)\eta(\boldsymbol{s}_2) = \begin{pmatrix} \cos\theta_p & \sin\theta_p e^{-i\phi_p} \\ \sin\theta_p e^{i\phi_p} & -\cos\theta_p \end{pmatrix} \begin{pmatrix} -\cos\frac{\theta_2}{2} e^{-i\phi_2} \\ -\sin\frac{\theta_2}{2} \end{pmatrix} = -\cos\frac{\theta_2}{2} e^{-i\phi_2} \begin{pmatrix} \cos\theta_p \\ \sin\theta_p e^{i\phi_p} \end{pmatrix} - \sin\frac{\theta_2}{2} \begin{pmatrix} \sin\theta_p e^{-i\phi_p} \\ -\cos\theta_p \end{pmatrix}$$

$$
\begin{aligned}
\chi^\dagger(\vec{s}_1)\,(\hat{n}_p\cdot\boldsymbol{\sigma})\,\eta(\vec{s}_2) &= \left[\cos\frac{\theta_1}{2}\begin{pmatrix}1\\0\end{pmatrix} + \sin\frac{\theta_1}{2}e^{-i\phi_1}\begin{pmatrix}0\\1\end{pmatrix}\right] \\
&\times\left[-\cos\frac{\theta_2}{2}e^{-i\phi_2}\begin{pmatrix}\cos\theta_p\\ \sin\theta_p e^{i\phi_p}\end{pmatrix} - \sin\frac{\theta_2}{2}\begin{pmatrix}\sin\theta_p e^{-i\phi_p}\\ -\cos\theta_p\end{pmatrix}\right]^T \\
&= -\cos\frac{\theta_1}{2}\cos\frac{\theta_2}{2}e^{-i\phi_2}\cos\theta_p - \cos\frac{\theta_1}{2}\sin\frac{\theta_2}{2}\sin\theta_p e^{-i\phi_p} \\
&-\sin\frac{\theta_1}{2}\cos\frac{\theta_2}{2}e^{-i(\phi_1+\phi_2)}\sin\theta_p e^{i\phi_p} + \sin\frac{\theta_1}{2}e^{-i\phi_1}\sin\frac{\theta_2}{2}\cos\theta_p
\end{aligned}
$$

4 Calculating integral in spherical coordinates

$$
\begin{aligned}
\int d\hat{n}\chi^\dagger(\vec{s}_1)(\hat{n}\cdot\boldsymbol{\sigma})\eta(\vec{s}_2) = \int_0^{2\pi} d\phi_p \int_0^{\pi} d\theta_p \sin\theta_p e^{i\vec{p}\cdot\vec{r}} \times \\
\Big[-\cos\frac{\theta_1}{2}\cos\frac{\theta_2}{2}e^{-i\phi_2}\cos\theta_p - \cos\frac{\theta_1}{2}\sin\frac{\theta_2}{2}\sin\theta_p e^{-i\phi_p} \\
-\sin\frac{\theta_1}{2}\cos\frac{\theta_2}{2}e^{-i(\phi_1+\phi_2)}\sin\theta_p e^{\phi_p} + \sin\frac{\theta_1}{2}e^{-i\phi_1}\sin\frac{\theta_2}{2}\cos\theta_p\Big]
\end{aligned}
$$

Some expressions for spherical harmonics

$$
Y_0^0 = \frac{1}{2}\sqrt{\frac{1}{\pi}} \qquad Y_1^1 = -\frac{1}{2}\sqrt{\frac{3}{2\pi}}\sin\theta e^{i\phi} \tag{1}
$$

$$
Y_1^0 = \frac{1}{2}\sqrt{\frac{3}{\pi}}\cos\theta \qquad Y_1^{-1} = \frac{1}{2}\sqrt{\frac{3}{2\pi}}\sin\theta e^{-i\phi} \tag{2}
$$

$$
Y_\ell^{m*}(\theta,\varphi) = (-1)^m Y_\ell^{-m}(\theta,\varphi) \tag{3}
$$

Using the expression for $e^{i\vec{p}\vec{r}}$ and substituting a trigonometric function for the corresponding spherical harmonic

$$
\begin{aligned}
&\int_0^{2\pi} d\phi_p \int_0^{\pi} d\theta_p \sin\theta_p e^{i\vec{p}\cdot\vec{r}} \sum_{l=0}^{\infty}\sum_{m=-l}^{l} 4\pi i^l j_l(kr) Y_l^m(\theta_p,\varphi_p) Y_l^{m*}(\theta_r,\varphi_r) \times \\
&\Big[-\cos\frac{\theta_1}{2}\cos\frac{\theta_2}{2}e^{-i\phi_2}\cdot 2\sqrt{\frac{\pi}{3}}Y_1^0(\theta_p,\phi_p) - \cos\frac{\theta_1}{2}\sin\frac{\theta_2}{2}\cdot 2\sqrt{\frac{2\pi}{3}}Y_1^{-1}(\theta_p,\phi_p) \\
&+\sin\frac{\theta_1}{2}\cos\frac{\theta_2}{2}e^{-i(\phi_1+\phi_2)}\cdot 2\sqrt{\frac{2\pi}{3}}Y_1^1(\theta_p,\phi_p) + \sin\frac{\theta_1}{2}e^{-i\phi_1}\sin\frac{\theta_2}{2}\cdot 2\sqrt{\frac{\pi}{3}}Y_1^0(\theta_p,\phi_p)\Big]
\end{aligned}
$$

Applying orthogonality $\int_{\theta=0}^{\pi}\int_{\varphi=0}^{2\pi} Y_\ell^m\, Y_{\ell'}^{m'*}\, d\Omega = \delta_{\ell\ell'}\,\delta_{mm'}$, (**MISTAKE** I forgot that orthogonality uses $Y_1^{-1*}(\theta_p, \varphi_p)$ instead of $Y_1^{-1}(\theta_p, \varphi_p)$. Must add corresponding $(-1)^m$ to each term)

$$
\begin{aligned}
4\pi j_1(kr) \int_0^{2\pi} d\phi_p \int_0^{\pi} d\theta_p \sin\theta_p \Bigg[& -2\sqrt{\frac{\pi}{3}} \cos\frac{\theta_1}{2} \cos\frac{\theta_2}{2} e^{-i\phi_2} \\
& \times \underbrace{Y_1^0(\theta_p, \varphi_p) Y_1^{0*}(\theta_p, \varphi_p)}_{=1} Y_1^{0*}(\theta_r, \varphi_r) \\
& + (-2)\sqrt{\frac{2\pi}{3}} \cos\frac{\theta_1}{2} \sin\frac{\theta_2}{2} \\
& \times \underbrace{Y_1^{-1}(\theta_p, \varphi_p) Y_1^{-1*}(\theta_p, \varphi_p)}_{=1} Y_1^{-1*}(\theta_r, \varphi_r) \\
& + 2\sqrt{\frac{2\pi}{3}} \sin\frac{\theta_1}{2} \cos\frac{\theta_2}{2} e^{-i(\phi_1+\phi_2)} \\
& \times \underbrace{Y_1^1(\theta_p, \varphi_p) Y_1^{1*}(\theta_p, \varphi_p)}_{=1} Y_1^{1*}(\theta_r, \varphi_r) \\
& + 2\sqrt{\frac{\pi}{3}} \sin\frac{\theta_1}{2} \sin\frac{\theta_2}{2} e^{-i\phi_1} \\
& \times \underbrace{Y_1^0(\theta_p, \varphi_p) Y_1^{0*}(\theta_p, \varphi_p)}_{=1} Y_1^{0*}(\theta_r, \varphi_r) \Bigg]
\end{aligned}
$$

5 Using the dot products

$$
\begin{aligned}
\psi(\theta_r, \phi_r; \theta_1, \phi_1; \theta_2, \phi_2) \propto \Bigg[& \left(-2\sqrt{\frac{2\pi}{3}} \cos\frac{\theta_1}{2} \sin\frac{\theta_2}{2} \right) Y_1^{*-1}(\theta_r, \phi_r) \\
& + \left(-2\sqrt{\frac{\pi}{3}} \cos\frac{\theta_1}{2} \cos\frac{\theta_2}{2} e^{-i\phi_2} + 2\sqrt{\frac{\pi}{3}} \sin\frac{\theta_1}{2} \sin\frac{\theta_2}{2} e^{-i\phi_1} \right) Y_1^{*0}(\theta_r, \phi_r) \\
& + \left(2\sqrt{\frac{2\pi}{3}} \sin\frac{\theta_1}{2} \cos\frac{\theta_2}{2} e^{-i(\phi_1+\phi_2)} \right) Y_1^{*1}(\theta_r, \phi_r) \Bigg]
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{old}(\theta_r, \phi_r; \theta_1, \phi_1; \theta_2, \phi_2) \propto \Bigg[& \left(-2\sqrt{\frac{\pi}{3}} \cos\frac{\theta_1}{2} \sin\frac{\theta_2}{2} e^{i\phi_2} \right) Y_1^{*-1}(\theta_r, \phi_r) \\
& + \left(2\sqrt{\frac{\pi}{3}} \cos\frac{\theta_1}{2} \cos\frac{\theta_2}{2} - 2\sqrt{\frac{\pi}{3}} \sin\frac{\theta_1}{2} \sin\frac{\theta_2}{2} e^{i(\phi_2-\phi_1)} \right) Y_1^{*0}(\theta_r, \phi_r) \\
& + \left(2\sqrt{\frac{2\pi}{3}} \sin\frac{\theta_1}{2} \cos\frac{\theta_2}{2} e^{-i\phi_1} \right) Y_1^{*1}(\theta_r, \phi_r) \Bigg]
\end{aligned}
$$

The goal is to express the wave function in a compact vector form. The wave function proportional to a sum over spherical harmonics could be written as a dot product involving vector $\vec{C}$

$$\psi \propto j_1(pr)\left(\vec{C}\cdot\hat{r}\right)$$

Vector $\vec{C}$ depends on the spin orientation of the fermions, $\vec{s}_1$ and $\vec{s}_2$. We start with the components of $\vec{C}$ derived from the matrix element calculation.

5.1 Initial expression for coefficients

In spherical basis

$$Y_1^0(\hat{r}) = \frac{3}{4\pi}n_z$$
$$Y_1^{\pm 1}(\hat{r}) = \mp\sqrt{\frac{3}{8\pi}}(n_x \pm i n_y)$$

where $\hat{r} = (n_x, n_y, n_z)$

Rewriting expression as a dot product

$$\sum_m C_m Y_1^{m*} = \sqrt{\frac{3}{4\pi}}\vec{C}\cdot\vec{t}$$
$$\sum_m C_m Y_1^{m*} = A\cdot Y_1^{*-1} + B\cdot Y_1^{*0} + C\cdot Y_1^{*1} =$$
$$= A\left(-\sqrt{\frac{3}{4\pi}}(n_x - i n_y)\right) + B\left(-\sqrt{\frac{3}{4\pi}}n_z\right) + C\left(-\sqrt{\frac{3}{4\pi}}(n_x + i n_y)\right)$$

Group around $\hat{r}$ components

$$\sum_m C_m Y_1^{m*} = \sqrt{\frac{3}{4\pi}}\left(-(A+C)n_x + i(A-C)n_y - Bn_z\right)$$

$$C_x = -(A+C) = 2\sqrt{\frac{\pi}{3}}\cos\frac{\theta_1}{2}\sin\frac{\theta_2}{2}e^{i\phi_2} - 2\sqrt{\frac{\pi}{3}}\sin\frac{\theta_1}{2}\cos\frac{\theta_2}{2}e^{-i\phi_1}$$
$$C_y = i(A-C) = -2i\sqrt{\frac{\pi}{3}}\cos\frac{\theta_1}{2}\sin\frac{\theta_2}{2}e^{i\phi_2} - 2i\sqrt{\frac{2\pi}{3}}\sin\frac{\theta_1}{2}\cos\frac{\theta_2}{2}e^{-i\phi_1}$$
$$C_z = -B = -2\sqrt{\frac{\pi}{3}}\cos\frac{\theta_1}{2}\cos\frac{\theta_2}{2} + 2\sqrt{\frac{\pi}{3}}\sin\frac{\theta_1}{2}\sin\frac{\theta_2}{2}e^{i(\phi_2-\phi_1)}$$

From the integral over the momentum direction of the decaying particles. we get the following complex coefficients:

$$
\begin{aligned}
C_x &= 2\sqrt{\frac{\pi}{3}}\left(c_1 s_2 e^{i\phi_2} - s_1 c_2 e^{-i\phi_1}\right)\\
C_y &= -2i\sqrt{\frac{\pi}{3}}\left(c_1 s_2 e^{i\phi_2} + s_1 c_2 e^{-i\phi_1}\right)\\
C_z &= -2\sqrt{\frac{\pi}{3}}\left(c_1 c_2 - s_1 s_2 e^{i(\phi_2-\phi_1)}\right)
\end{aligned}
$$

$$
\begin{aligned}
\vec{C} &= \alpha\vec{s}_1 + \beta\vec{s}_2 + \gamma(\vec{s}_1\times\vec{s}_2)\\
C_x &= 2\sqrt{\frac{\pi}{3}}\cos\frac{\theta_1}{2}\sin\frac{\theta_2}{2}e^{i\phi_2} - 2\sqrt{\frac{\pi}{3}}\sin\frac{\theta_1}{2}\cos\frac{\theta_2}{2}e^{-i\phi_1} =\\
&= \alpha\left(\sin\theta_1\cos\phi_1\right) + \beta\left(\sin\theta_2\cos\phi_2\right) + \gamma(s_{1x}s_{2z} - s_{1z}s_{2y})\\
&= \alpha\left(\sin\theta_1\cos\phi_1\right) + \beta\left(\sin\theta_2\cos\phi_2\right) + \gamma(\sin\theta_1\cos\phi_1\cos\theta_2 - \cos\theta_1\sin\theta_2\sin\phi_2)\\
C_y &= -2i\sqrt{\frac{\pi}{3}}\cos\frac{\theta_1}{2}\sin\frac{\theta_2}{2}e^{i\phi_2} - 2i\sqrt{\frac{2\pi}{3}}\sin\frac{\theta_1}{2}\cos\frac{\theta_2}{2}e^{-i\phi_1} =\\
&= \alpha\left(\sin\theta_1\sin\phi_1\right) + \beta\left(\sin\theta_2\sin\phi_2\right) + \gamma(s_{1z}s_{2x} - s_{1x}s_{2z})\\
&= \alpha\left(\sin\theta_1\sin\phi_1\right) + \beta\left(\sin\theta_2\cos\phi_2\right) + \gamma(\cos\theta_1\sin\theta_2\cos\phi_2 - \sin\theta_1\cos\phi_1\cos\theta_2)\\
C_z &= -2\sqrt{\frac{\pi}{3}}\cos\frac{\theta_1}{2}\cos\frac{\theta_2}{2} + 2\sqrt{\frac{\pi}{3}}\sin\frac{\theta_1}{2}\sin\frac{\theta_2}{2}e^{i(\phi_2-\phi_1)} =\\
&= \alpha\cos\theta_1 + \beta\cos\theta_2 + \gamma(s_{1x}s_{2y} - s_{1y}s_{2x})\\
&= \alpha\cos\theta_1 + \beta\cos\theta_2 + \gamma(\sin\theta_1\cos\phi_1\sin\theta_2\sin\phi_2 - \sin\theta_1\sin\phi_1\sin\theta_2\cos\phi_2)
\end{aligned}
$$

Here, $c_i = \cos(\theta_i/2)$ and $s_i = \sin(\theta_i/2)$

5.2 Expressing coefficients using spin vector components

The components of the spin vectors are:

$$\vec{s}_i = (\sin\theta_i\cos\phi_i, \sin\theta_i\sin\phi_i, \cos\theta_i) = (s_{ix}, s_{iy}, s_{iz})$$

Some useful half-angle identities:

$$c_i^2 = \frac{1+s_{iz}}{2}, \quad s_i^2 = \frac{1-s_{iz}}{2}, \quad 2s_ic_i = \sqrt{1-s_{iz}^2} = \sin\theta_i$$

Using these, we can express trigonometric functions of the angles:

$$\cos\phi_i = \frac{s_{ix}}{\sin\theta_i} = \frac{s_{ix}}{2s_ic_i}, \quad \sin\phi_i = \frac{s_{iy}}{\sin\theta_i} = \frac{s_{iy}}{2s_ic_i}$$

5.2.1 Separating real and imaginary parts

We expand the complex exponent $e^{i\phi} = \cos\phi + i\sin\phi$ and separate the real and imaginary parts of each coefficient.

z-components: From equation (?), the imaginary part of C_z is:

$$\mathrm{Im}(C_x) = 2\sqrt{\frac{\pi}{3}} s_1 s_2 \sin(\phi_2 - \phi_1)$$

$$\vec{s}_1 \times \vec{s}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ s_{1x} & s_{1y} & s_{1z} \\ s_{2x} & s_{2y} & s_{2z} \end{vmatrix} = \vec{i}(s_{1y}s_{2z} - s_{1z}s_{2y}) - \vec{j}(s_{1x}s_{2z} - s_{1z}s_{2x}) + \vec{k}(s_{1x}s_{2y} - s_{1y}s_{2x})$$

z-component of the cross product $\vec{s}_1 \times \vec{s}_2$ is:

$$\begin{aligned}(\vec{s}_1 \times \vec{s}_2)_z &= s_{1x}s_{2y} - s_{1y}s_{2x} \\ &= (\sin\theta_1 \cos\phi_1)(\sin\theta_2 \sin\phi_2) - (\sin\theta_1 \sin\phi_1)(\sin\theta_2 \cos\phi_2) \\ &= \sin\theta_1 \sin\theta_2 \,(\cos\phi_1 \sin\phi_2 - \sin\phi_1 \cos\phi_2) \\ &= \sin\theta_1 \sin\theta_2 \sin(\phi_2 - \phi_1)\end{aligned}$$

Using the identity $\sin\theta_i = 2s_i c_i$, we get:

$$(\vec{s}_1 \times \vec{s}_2)_z = (2s_1c_1)\,(2s_1c_1) \sin(\phi_2 - \phi_1)$$

Now, we can express term form $\mathrm{Im}(C_z)$:

$$s_1 s_2 \sin(\phi_2 - \phi_1) = \frac{(\vec{s}_1 \times \vec{s}_2)_z}{4c_1c_2}$$

Subsitute back into the $\mathrm{Im}(C_z)$:

$$\mathrm{Im}(C_z) = 2\sqrt{\frac{\pi}{3}} \frac{(\vec{s}_1 \times \vec{s}_2)_z}{4c_1c_2} = \frac{\sqrt{\pi/3}}{2c_1c_2}(\vec{s}_1 \times \vec{s}_2)_z$$

x- and y-components: Following a similar procedure for C_x and C_y:

$$\mathrm{Im}(C_x) = 2\sqrt{\frac{\pi}{3}}\,(c_1 s_2 \sin\phi_2 + s_1 c_2 \sin\phi_1) = \frac{\sqrt{\pi/3}}{c_1c_2}\left(c_1^2 s_{2y} + c_2^2 s_{1y}\right)$$

$$\mathrm{Im}(C_y) = -\mathrm{Re}\left(2\sqrt{\frac{\pi}{3}}(c_1 s_2 e^{i\phi_2} + s_1 c_2 e^{-i\phi_1})\right) = -2\sqrt{\frac{\pi}{3}}(c_1 s_2 \cos\phi_2 + s_1 c_2 \cos\phi_1) = -\frac{\sqrt{\pi/3}}{c_1c_2}(c_1^2 s_{2x} + c_2^2 s_{1x})$$

Replacing $c_i^2 = (1 + s_{iz})/2$, we get:

$$\begin{aligned}
\mathrm{Im}(C_x) &= \frac{\sqrt{\pi/3}}{2c_1c_2}\left[(1+s_{1z})s_{2y} + (1+s_{2z})s_{1y}\right] \\
\mathrm{Im}(C_y) &= -\frac{\sqrt{\pi/3}}{2c_1c_2}\left[(1+s_{1z})s_{2x} + (1+s_{2z})s_{1x}\right]
\end{aligned}$$

The terms in the brackets can be expanded:

$$\begin{aligned}
(1+s_{1z})s_{2y} + (1+s_{2z})s_{1y} &= s_{2y} + s_{1y} + s_{1z}s_{2y} + s_{1y}s_{2z} \\
&= s_{1y} + s_{2y} + (s_{1y}s_{2z} - s_{1z}s_{2y}) + 2s_{1z}s_{2y} \\
&= s_{1y} + s_{2y} + (\vec{s}_1 \times \vec{s}_2)_x + 2s_{1z}s_{2y} \\
(1+s_{1z})s_{2x} + (1+s_{2z})s_{1x} &= s_{2x} + s_{1x} + s_{1z}s_{2x} + s_{2z}s_{1x} \\
&= s_{1x} + s_{2x} + (s_{1z}s_{2x} - s_{1x}s_{2z}) + 2s_{1x}s_{2z} \\
&= s_{1x} + s_{2x} + (\vec{s}_1 \times \vec{s}_2)_y + 2s_{1x}s_{2z}
\end{aligned}$$

5.2.2 Analysis of $\ell = 1$ contributions

The wave function's dependence on the direction $\hat{r}$ comes from terms with $\ell = 1$, which behave like vectors. All terms derived from the expansion are vectorial and contribute to $\vec{C}_{\ell=1}$. There are no scalar ($\ell = 0$) contributions like $\vec{s}_1 \cdot \vec{s}_2$.

The components of the imaginary part of $\vec{C}$ can be grouped into distinct physical contributions:

$$\begin{aligned}
\mathrm{Im}(C_x) &= \frac{\sqrt{\pi/3}}{2c_1c_2}\left[\underbrace{(s_{1y} + s_{2y})}_{\text{Sum of spins}} + \underbrace{(\vec{s}_1 \times \vec{s}_2)_x}_{\text{Spin correlation}} + \underbrace{2s_{1z}s_{2y}}_{\text{Higher-order}}\right] \\
\mathrm{Im}(C_y) &= -\frac{\sqrt{\pi/3}}{2c_1c_2}\left[\underbrace{(s_{1x} + s_{2x})}_{\text{Sum of spins}} + \underbrace{(\vec{s}_1 \times \vec{s}_2)_y}_{\text{Spin correlation}} + \underbrace{2s_{1x}s_{2z}}_{\text{Higher-order}}\right] \\
\mathrm{Im}(C_z) &= \frac{\sqrt{\pi/3}}{2c_1c_2}\left[\underbrace{(\vec{s}_1 \times \vec{s}_2)_z}_{\text{Spin correlation}}\right]
\end{aligned}$$

The complete expression for the imaginary part of $\vec{C}$ is:

$$\mathrm{Im}(C_x) = \frac{\sqrt{\pi/3}}{2c_1c_2}\left[(s_{1y} + s_{2y}) + (\vec{s}_1 \times \vec{s}_2)_x + 2s_{1z}s_{2y}\right]$$
$$\mathrm{Im}(C_y) = -\frac{\sqrt{\pi/3}}{2c_1c_2}\left[(s_{1x} + s_{2x}) + (\vec{s}_1 \times \vec{s}_2)_y + 2s_{1x}s_{2z}\right]$$
$$\mathrm{Im}(C_z) = \frac{\sqrt{\pi/3}}{2c_1c_2}(\vec{s}_1 \times \vec{s}_2)_z$$

6 Conclusion